GRAPH THEORY

From Matching to Network Flows

Lecture 14 • Real-life modeling • Ford–Fulkerson with worked examples

Learning goal: model, calculate, explain, and verify maximum flow.

Learning outcomes

By the end of this lecture, students should be able to:

- Recognize when a real problem can be represented as a flow network.
- Apply capacity constraints and flow conservation.
- Find augmenting paths and update residual capacities.
- Execute Ford–Fulkerson on a small network and justify optimality.

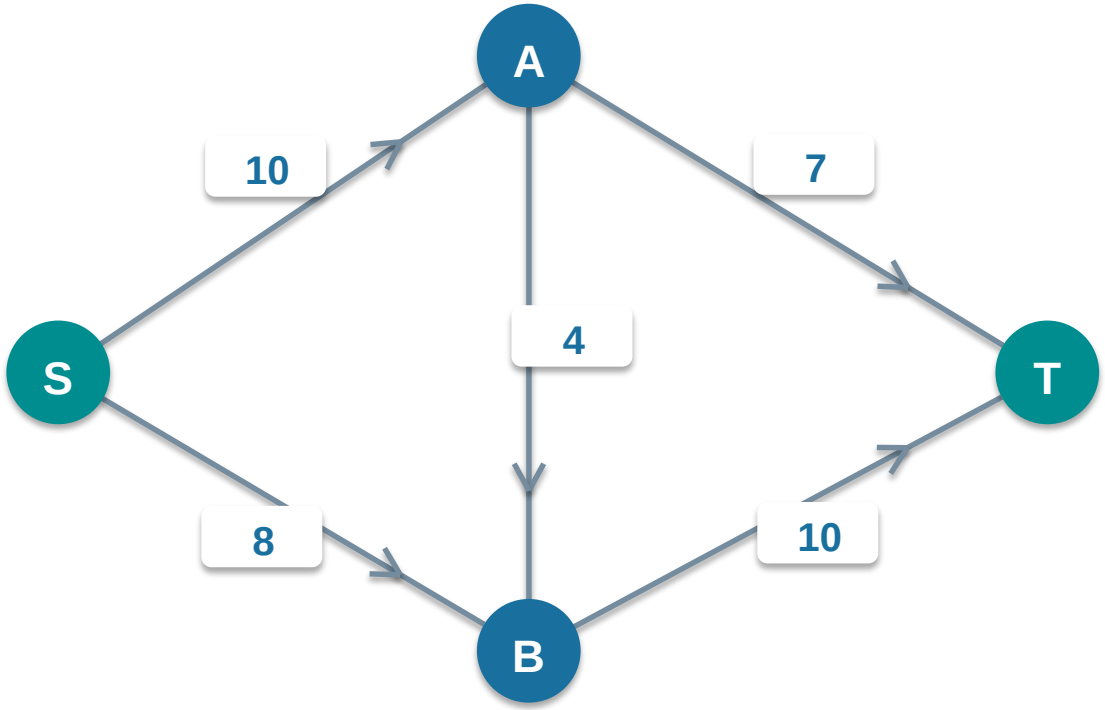
Real problem: campus internet backbone

Scenario: the main server must deliver data to a remote campus network.

- Links have different bandwidth limits (Gbps).
- Intermediate switches can forward traffic but cannot create data.
- Some links may become bottlenecks even when the server has spare bandwidth.
- Question: What is the maximum sustainable throughput from S to T?

Translate the situation into a directed graph

Vertices = routers/switches; directed edges = allowed data routes; labels = link capacities.



Think before calculating

S can send at most 18 Gbps and T can receive at most 17 Gbps. Which bound is achievable?

Assumption: capacities are independent, directed, and measured in Gbps.

Three rules for a feasible flow

1 Capacity

For every edge: 0 ≤ flow ≤ capacity.

2 Conservation

At each intermediate vertex: inflow = outflow.

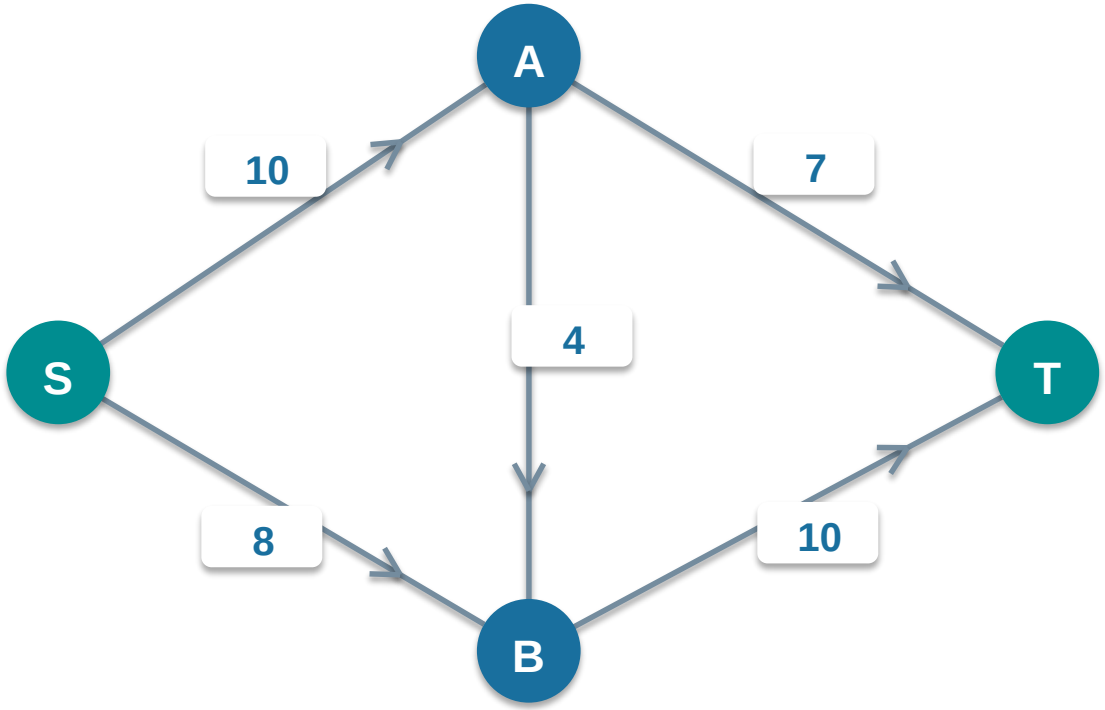
3 Flow value

Total flow leaving S equals total flow entering T.

Example at A: f(S,A) = f(A,T) + f(A,B).

Capacity is an upper limit, not the amount that must be used.

Why adding link capacities is not enough



Upper bounds

Out of S: 10 + 8 = 18 Gbps

Into T: 7 + 10 = 17 Gbps

Therefore max flow ≤ 17 Gbps.

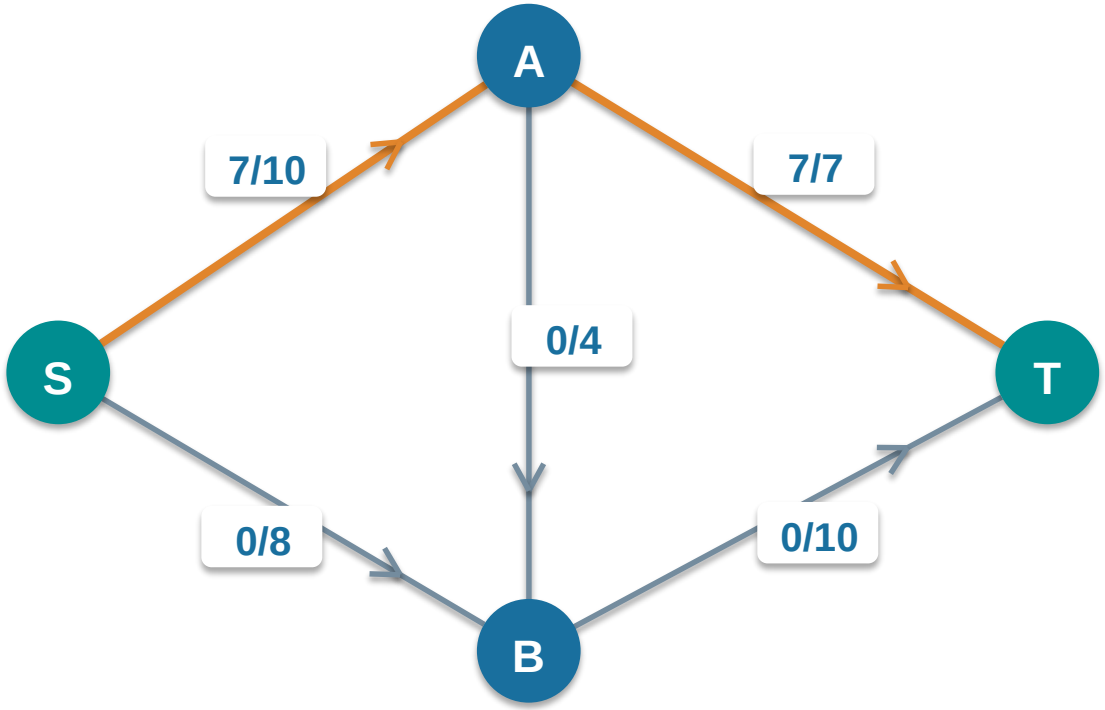
Can we construct a feasible flow of 17? If yes, the bound is tight.

Ford–Fulkerson: the algorithmic idea

- Start with zero flow on every edge.
- Build the residual network: what additional flow can still be sent?
- Find an S→T path whose residual capacities are positive.
- Push the path bottleneck (smallest residual capacity).
- Update forward and reverse residual capacities; repeat until no path remains.

Worked example — Step 1: send 7 Gbps

Choose path S → A → T; bottleneck = min(10, 7) = 7.



Current total flow

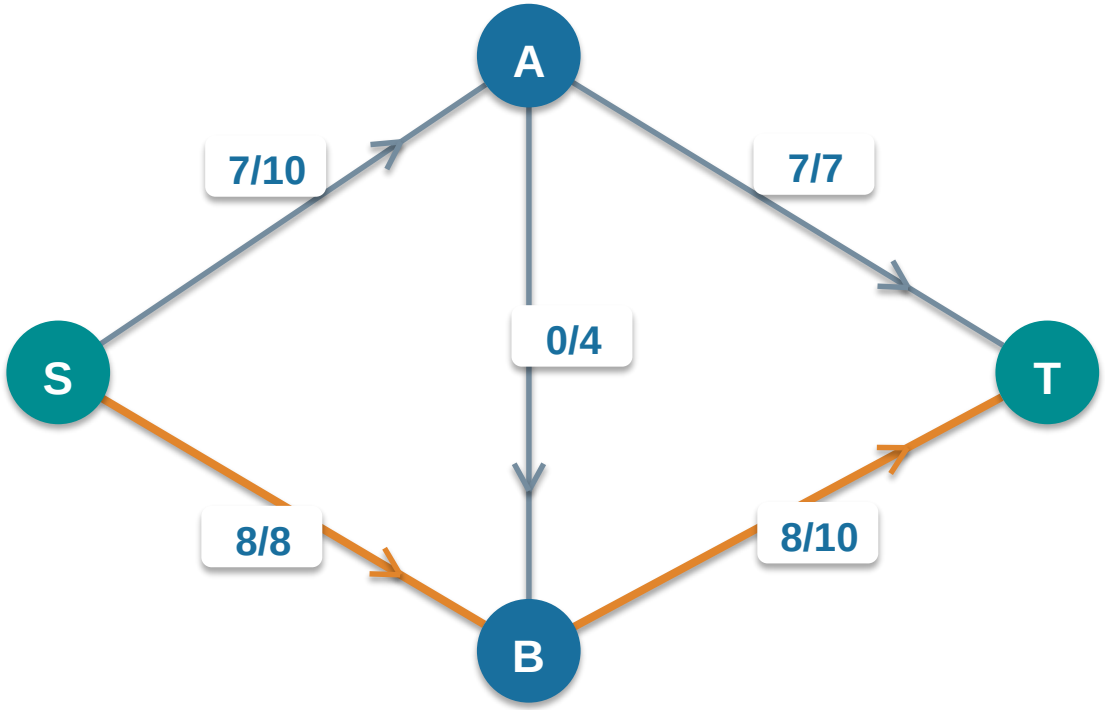
7 Gbps reaches T.

Residual on S→A = 3.

Residual on A→T = 0.

Worked example — Step 2: send 8 Gbps

Choose path S → B → T; bottleneck = min(8, 10) = 8.



Current total flow

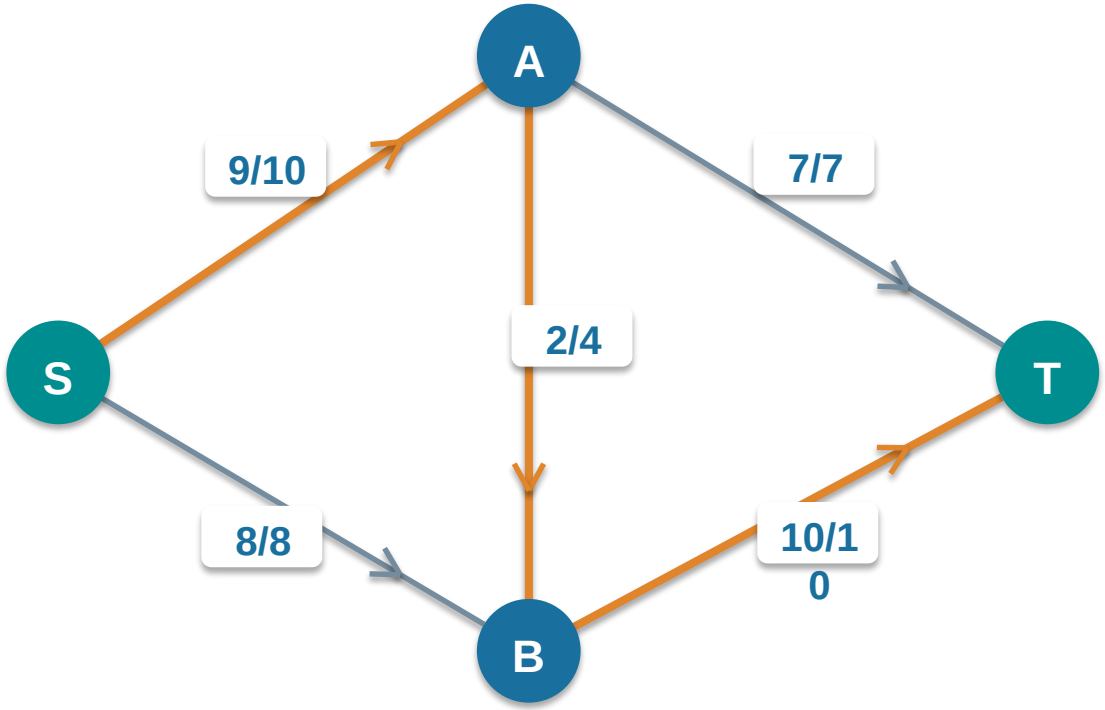
7 + 8 = 15 Gbps.

S→B is saturated.

B→T has 2 Gbps available.

Worked example — Step 3: send 2 Gbps

Choose path S → A → B → T; residual bottleneck = min(3, 4, 2) = 2.

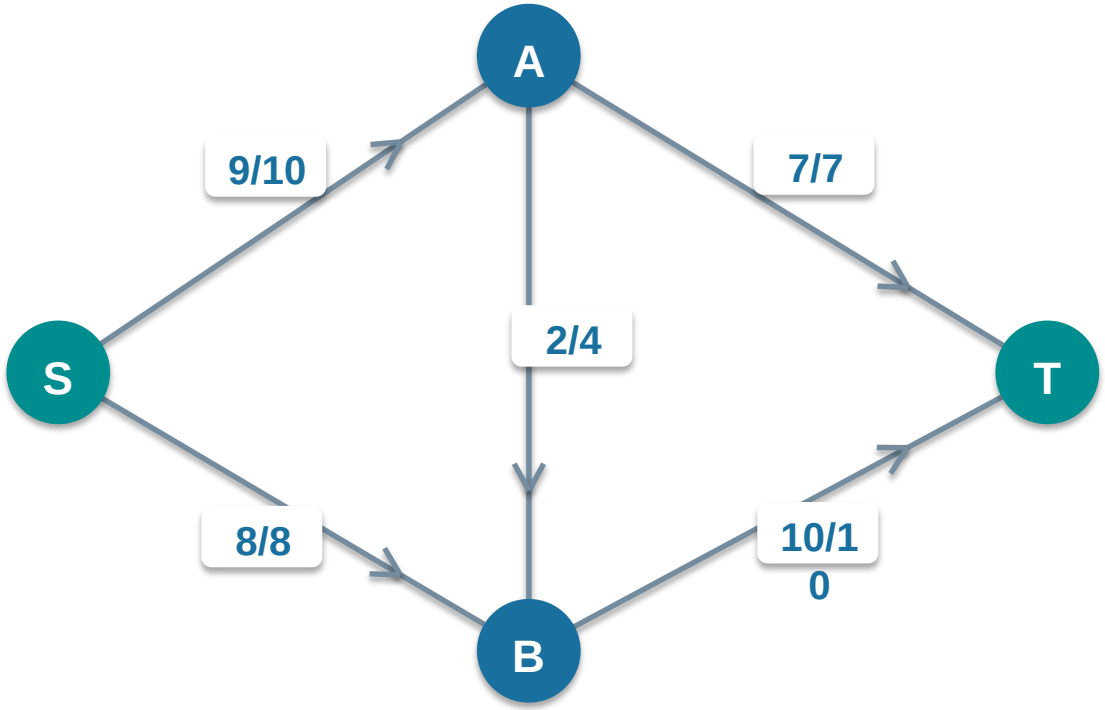


Current total flow

15 + 2 = 17 Gbps.

The incoming links to T are now saturated.

Verify the solution: 17 Gbps is maximum



Conservation check

At A: 9 = 7 + 2

At B: 8 + 2 = 10

At T: 7 + 10 = 17

Optimality certificate

The cut {S,A,B} | {T} has capacity 7 + 10 = 17. A feasible flow of 17 meets this upper bound.

What exactly is a residual network?

Forward residual edge

If capacity c and current flow f, remaining forward capacity is c − f.

Reverse residual edge

If current flow is f, the residual network allows up to f units to be sent backward, canceling previously assigned flow.

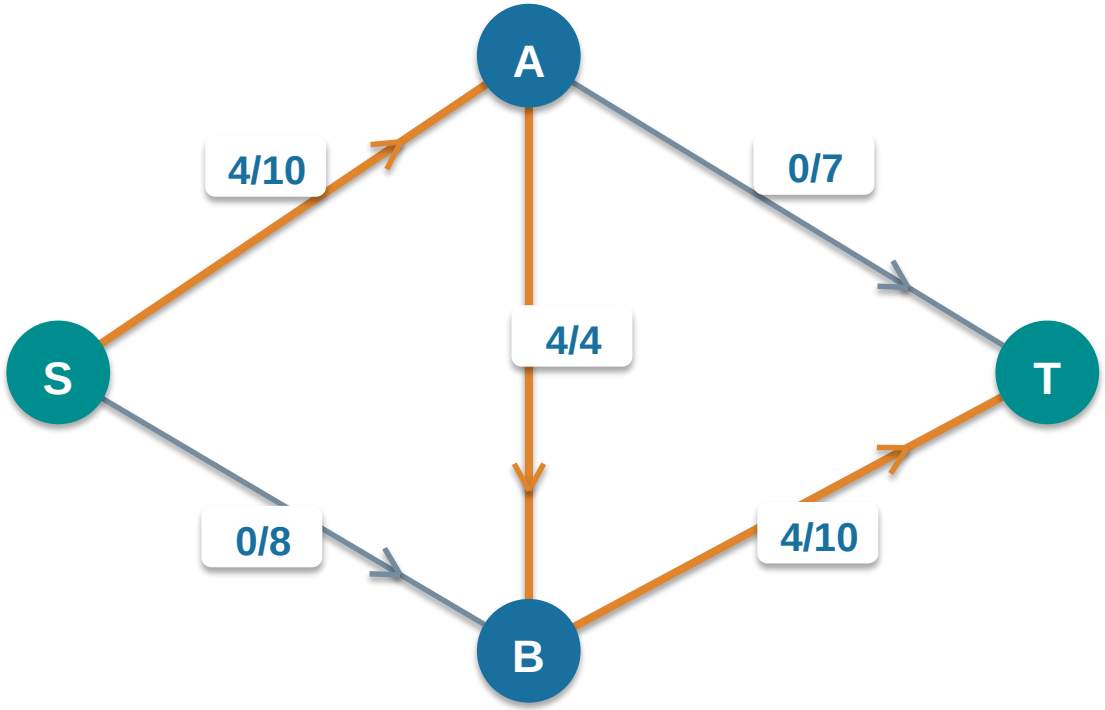
Example: if A→B has capacity 4 and flow 3:

Residual A→B = 1; residual B→A = 3.

Reverse residual edges do not mean the physical network acquired a new reverse link.

Worked example — A poor first path

Choose S→A→B→T first, pushing 4 Gbps.



Flow after first path

S→A = 4

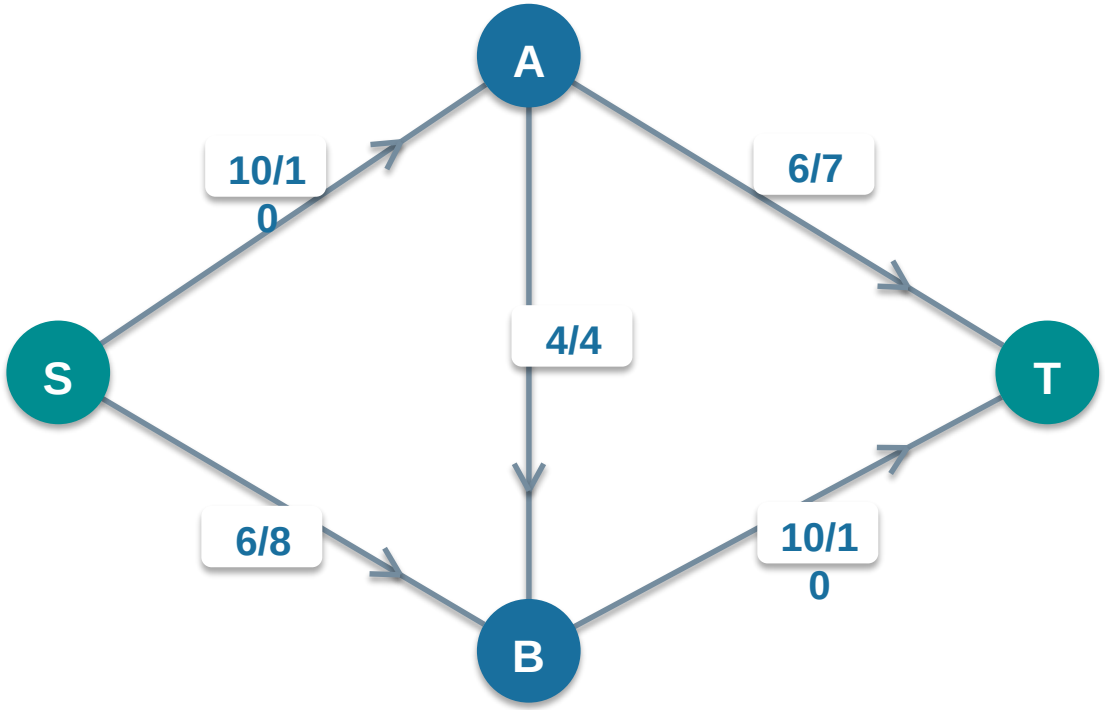
A→B = 4

B→T = 4

Total = 4 Gbps

Continue the poor-path sequence

Then send 6 on S→A→T, and 6 on S→B→T.



Current total = 16

A→T still has 1 Gbps available.

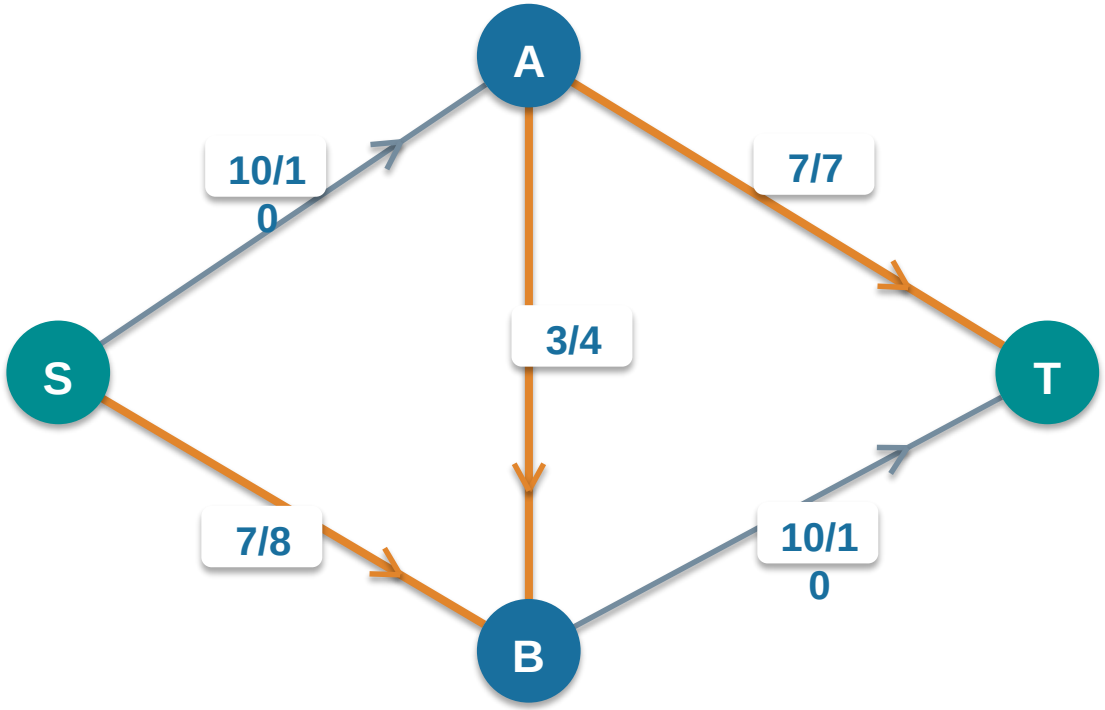
S→B still has 2 Gbps available.

But A→B is saturated.

Question: Is 16 maximum, or can we rearrange the existing flow?

Reverse residual edge unlocks the last 1 Gbps

Use residual path S→B→A→T; B→A reverses 1 unit of earlier A→B flow.



Flow adjustment

S→B: 6 → 7

A→B: 4 → 3 (cancel 1)

A→T: 6 → 7

Total: 16 → 17 Gbps

Key lesson: a greedy choice can be repaired without restarting the entire solution.

Ford–Fulkerson: algorithm trace

Capacities and path choices for the same campus network.

Initial	—	**0**
1	S–A–T; +7	**7**
2	S–B–T; +8	**15**
3	S–A–B–T; +2	**17**
Alternative 1	S–A–B–T; +4	**4**
Alternative 2	S–A–T; +6	**10**
Alternative 3	S–B–T; +6	**16**
Alternative 4	S–B–A–T; +1 (reverse B–A)	**17**

Second real-world application: emergency supply routing

Problem

A disaster-response depot must deliver supplies through roads with daily vehicle-throughput limits. Multiple roads may share bottleneck intersections.

Graph model

Vertices = depots/intersections; directed edges = usable road directions; capacity = maximum trucks per day.

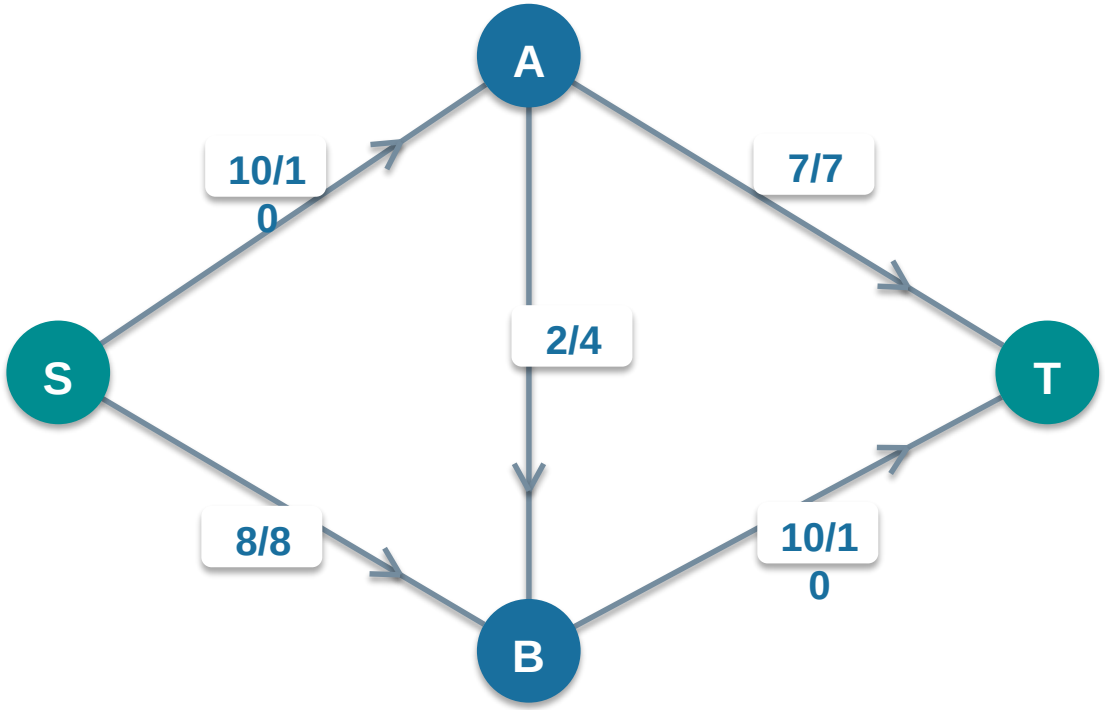
What is different from a shortest-path problem?

Shortest path minimizes cost for one route; maximum flow maximizes total throughput across many routes.

When does a flow model NOT fit?

- If a single indivisible object must follow one route, ordinary fractional flow may be inappropriate.
- If each edge has travel time and the goal is earliest arrival, maximum flow alone does not solve it.
- If several products compete for the same links, this becomes a multi-commodity problem.
- Always define the source, sink, capacity units, and conservation assumptions.

In-class problem: diagnose the claimed flow



Student claim

“This network carries 18 Gbps because all links leaving S are saturated.”

Tasks

1. Check flow conservation at A and B.
2. Explain why the claimed flow is infeasible.
3. State the actual maximum and justify it.

Quick conceptual checks

- Why can two different augmenting-path orders reach the same maximum flow?
- What does a reverse residual edge represent in physical terms?
- If a flow saturates all edges entering T, is it necessarily maximum? Explain.
- What is the difference between a feasible flow and a maximum flow?

Wrap-up and next lecture

Today

Matching → augmenting paths → capacities and conservation → Ford–Fulkerson → Edmond Karp → verified maximum flow.

Next lecture

Minimum cuts, Max-Flow Min-Cut Theorem, Edmonds–Karp, and how bipartite matching can be represented as a flow network.